

Physical Chemistry–III (CHB-504)

Unit 2: Colligative Properties

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We all know colligative is easy peasy

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Introduction: What Are Colligative Properties?

Definition: Colligative properties are those properties of a dilute solution that depend *only on the number of solute particles* (moles) dissolved in a given amount of solvent, and **not** on the chemical identity or nature of the solute.

The four colligative properties are:

1. Lowering of vapour pressure (Raoult's law)
2. Elevation of boiling point (ΔT_b)
3. Depression of freezing point (ΔT_f)
4. Osmotic pressure (π)

Why are they colligative? Each property arises from the change in chemical potential of the *solvent* caused by the presence of solute. Because:

$$\mu_A(\text{solution}) = \mu_A^*(\text{pure}) + RT \ln x_A$$

and $\ln x_A < 0$ whenever solute is present (regardless of its identity), all four properties depend solely on x_A , i.e., the *number* of solute molecules, not their type.

Raoult's Law and Vapour Pressure Lowering

Statement of Raoult's Law

For an ideal solution, the partial vapour pressure p_A of the solvent above the solution equals the product of its mole fraction x_A and the vapour pressure of the pure solvent p_A^* :

$$p_A = x_A p_A^*$$

Vapour Pressure Lowering

Since $x_A = 1 - x_B$ (where x_B is the mole fraction of solute):

$$p_A = (1 - x_B)p_A^*$$

$$p_A^* - p_A = x_B p_A^*$$

$$\frac{p_A^* - p_A}{p_A^*} = x_B$$

The **relative lowering of vapour pressure** equals the mole fraction of the solute. This is a colligative property.

Henry's Law and Ideal-Dilute Solution

For the *solute* in an ideal-dilute solution (when $x_B \rightarrow 0$):

$$p_B = K_H x_B$$

where K_H is Henry's law constant. Unlike Raoult's law, $K_H \neq p_B^*$ in general.

Molecular basis: In Raoult's law, solvent molecules are surrounded mostly by like molecules. In Henry's law, solute molecules are surrounded entirely by solvent molecules – a very different environment.

Elevation of Boiling Point

Why Boiling Point is Elevated (Qualitative)

The normal boiling point is the temperature at which the vapour pressure of the liquid equals 1 atm (external pressure). Since the solute lowers the vapour pressure of the solvent (Raoult's law), the solution must be *heated to a higher temperature* to restore the vapour pressure to 1 atm. Hence, the boiling point is elevated.

Thermodynamic Derivation: ΔT_b is a Colligative Property

Condition for boiling: At the boiling point, the chemical potential of solvent in the liquid solution equals that in the vapour:

$$\mu_A^{\text{liq}}(T, p) = \mu_A^{\text{vap}}(T, p)$$

For the solution (ideal dilute):

$$\mu_A^{\text{liq}} = \mu_A^{*,\text{liq}} + RT \ln x_A$$

For the pure vapour:

$$\mu_A^{\text{vap}} = \mu_A^{*,\text{vap}}$$

At equilibrium:

$$\begin{aligned} \mu_A^{*,\text{liq}} + RT \ln x_A &= \mu_A^{*,\text{vap}} \\ RT \ln x_A &= \mu_A^{*,\text{vap}} - \mu_A^{*,\text{liq}} = \Delta_{\text{vap}} G_m^* \end{aligned}$$

Using $\Delta_{\text{vap}} G_m^* = \Delta_{\text{vap}} H_m^* - T \Delta_{\text{vap}} S_m^*$ and that at the *pure boiling point* T_b^* : $\Delta_{\text{vap}} G_m^* = 0$:

$$RT \ln x_A = \Delta_{\text{vap}} H_m^* \left(1 - \frac{T_b^*}{T} \right) \cdot (-1) \quad (\text{after rearranging})$$

More carefully: differentiating $\Delta_{\text{vap}} G_m^*/T$ with respect to T (Gibbs–Helmholtz):

$$RT \ln x_A = -\Delta_{\text{vap}} H_m^* \left(\frac{T - T_b^*}{T} \right)$$

$$R \ln x_A = -\frac{\Delta_{\text{vap}} H_m^*}{T} \cdot \frac{T - T_b^*}{T_b^*} \quad (\text{since } T \approx T_b^* \text{ for dilute solution})$$

Let $\Delta T_b = T - T_b^* > 0$. For dilute solution, $\ln x_A \approx \ln(1 - x_B) \approx -x_B$:

$$-R x_B \approx -\frac{\Delta_{\text{vap}} H_m^* \Delta T_b}{(T_b^*)^2}$$

$$\boxed{\Delta T_b = \frac{R(T_b^*)^2}{\Delta_{\text{vap}} H_m^*} x_B}$$

Expression in Terms of Molality

For a dilute solution, $x_B \approx n_B/n_A = n_B M_A / (1000)$ (where M_A is molar mass of solvent in g/mol), and molality $m = n_B \times 1000 / (n_A M_A) \times (1/\text{kg solvent})$:

$$x_B = \frac{m M_A}{1000}$$

Substituting:

$$\Delta T_b = \frac{R(T_b^*)^2 M_A}{1000 \Delta_{\text{vap}} H_m^*} \cdot m = K_b m$$

$$\boxed{\Delta T_b = K_b m}$$

where the **ebullioscopic constant** (molal boiling point elevation constant) is:

$$\boxed{K_b = \frac{R(T_b^*)^2 M_A}{1000 \Delta_{\text{vap}} H_m^*}}$$

Why This is Colligative

$\Delta T_b = K_b m$ depends only on the molality m (number of moles of solute per kg solvent) and the constant K_b (a property of the solvent alone). It does **not** depend on what the solute is – only how many particles are present.

Molar Mass Determination

Rearranging $\Delta T_b = K_b \cdot \frac{w_B \times 1000}{M_B \times w_A}$ (where w_B and w_A are masses in grams):

$$\boxed{M_B = \frac{K_b \times w_B \times 1000}{\Delta T_b \times w_A}}$$

This is how the molar mass of a non-volatile solute is determined experimentally.

Depression of Freezing Point

Why Freezing Point is Depressed (Qualitative)

The freezing point is the temperature at which the chemical potential of liquid solvent equals that of solid solvent. Since dissolving a solute lowers μ_A^{liq} (but not μ_A^{solid}), equilibrium is achieved at a *lower* temperature where μ_A^{solid} has also decreased enough to match the lowered μ_A^{liq} . Hence the freezing point is depressed.

Equivalently: a solution has lower vapour pressure than pure solvent, so its vapour pressure curve intersects the solid vapour pressure curve (which is unchanged) at a lower temperature.

Thermodynamic Derivation

By the same method as boiling point elevation, but now with fusion instead of vaporisation:

At the freezing point of the solution, $\mu_A^{\text{liq}} = \mu_A^{\text{solid}}$:

$$\mu_A^{*,\text{liq}} + RT \ln x_A = \mu_A^{*,\text{solid}}$$

$$RT \ln x_A = \mu_A^{*,\text{solid}} - \mu_A^{*,\text{liq}} = -\Delta_{\text{fus}} G_m^*$$

Applying the Gibbs–Helmholtz equation and the approximation $T \approx T_f^*$:

$$R \ln x_A = \frac{\Delta_{\text{fus}} H_m^*}{(T_f^*)^2} (T - T_f^*)$$

Since $T < T_f^*$, let $\Delta T_f = T_f^* - T > 0$ and $\ln x_A \approx -x_B$:

$$-R x_B = -\frac{\Delta_{\text{fus}} H_m^*}{(T_f^*)^2} \Delta T_f$$

$$\Delta T_f = \frac{R(T_f^*)^2}{\Delta_{\text{fus}} H_m^*} x_B = K_f m$$

where the **cryoscopic constant** (molal freezing point depression constant) is:

$$K_f = \frac{R(T_f^*)^2 M_A}{1000 \Delta_{\text{fus}} H_m^*}$$

Depression of Freezing Point is a Colligative Property

The argument is identical to the boiling point case: ΔT_f depends only on the solute molality m , not on the chemical identity of the solute.

Molar Mass Determination

$$M_B = \frac{K_f \times w_B \times 1000}{\Delta T_f \times w_A}$$

Worked Example

Problem: 100 g of a compound dissolved in 750 g CCl_4 lowered the freezing point by 10.5 K. Given $K_f(\text{CCl}_4) = 32 \text{ K kg mol}^{-1}$ (typical), find the molar mass.

$$M_B = \frac{K_f \times w_B \times 1000}{\Delta T_f \times w_A} = \frac{32 \times 100 \times 1000}{10.5 \times 750} = \frac{3,200,000}{7875} \approx \boxed{406 \text{ g mol}^{-1}}$$

Osmotic Pressure and Van't Hoff Equation

What is Osmosis?

Osmosis is the spontaneous flow of solvent molecules through a *semipermeable membrane* (which allows solvent but not solute to pass) from a region of *lower* solute concentration (lower chemical potential of solvent) to a region of *higher* solute concentration (higher chemical potential of solvent).

The flow continues until the chemical potentials on both sides are equal.

Why Does Solvent Flow Spontaneously?

The chemical potential of the solvent in the solution is:

$$\mu_A^{\text{soln}} = \mu_A^* + RT \ln x_A < \mu_A^* = \mu_A^{\text{pure}}$$

Since the pure solvent has higher μ_A , solvent flows into the solution to equalise μ_A .

Definition of Osmotic Pressure

The **osmotic pressure** π is the excess pressure that must be applied to the *solution* side to halt the flow of solvent through the semipermeable membrane. It is the pressure that equalises the chemical potentials on both sides.

Derivation of Van't Hoff Equation

At equilibrium, with osmotic pressure π applied to the solution:

$$\mu_A^*(\text{pure}, P) = \mu_A(\text{soln}, P + \pi)$$

The effect of pressure on chemical potential:

$$\mu_A(\text{soln}, P + \pi) = \mu_A(\text{soln}, P) + \int_P^{P+\pi} \bar{V}_A \, dP'$$

For incompressible solvent, $\bar{V}_A \approx V_A^*$ (molar volume of pure solvent):

$$\mu_A(\text{soln}, P + \pi) = \mu_A^* + RT \ln x_A + V_A^* \pi$$

Setting equal to $\mu_A^*(P)$:

$$\begin{aligned} \mu_A^* &= \mu_A^* + RT \ln x_A + V_A^* \pi \\ -RT \ln x_A &= V_A^* \pi \end{aligned}$$

For dilute solution: $\ln x_A = \ln(1 - x_B) \approx -x_B$, and $x_B \approx n_B/n_A$ (since $n_B \ll n_A$):

$$RT x_B = V_A^* \pi$$

$$RT \frac{n_B}{n_A} = V_A^* \pi$$

Since $n_A V_A^* = V$ (total volume of solution $\approx$ volume of solvent for dilute solution):

$$RT \frac{n_B}{V} = \pi$$

$$\boxed{\pi V = n_B RT}$$

or equivalently, with $c = n_B/V$ (molar concentration):

$$\boxed{\pi = cRT}$$

This is the **Van't Hoff equation** for osmotic pressure. Its mathematical form is analogous to the ideal gas law $PV = nRT$.

Why Osmosis is a Colligative Property

$\pi = cRT$ depends only on c (number of moles of solute per unit volume), not on the identity of the solute. Hence osmotic pressure is colligative.

Osmosis is a Colligative Property: Proof

Statement: Osmosis is a colligative property because the osmotic pressure depends only on the *molar concentration* (or number density) of solute particles, regardless of their chemical nature.

Evidence:

- 1 M glucose solution and 1 M sucrose solution have the same osmotic pressure at the same T .
- $\pi = cRT$: only c and T appear; no chemical identity terms.
- Derived from $-RT \ln x_A = \pi V_A^*$: only the mole fraction of solvent (hence number of solute particles) matters.

Van't Hoff Factor for Electrolytes

For electrolytes that dissociate, the effective number of particles is greater. Van't Hoff introduced the factor i :

$$\pi = i cRT, \quad \Delta T_b = i K_b m, \quad \Delta T_f = i K_f m$$

where i = (number of particles produced per formula unit). For NaCl: $i = 2$; for CaCl_2 : $i = 3$, etc.

Determining Molar Mass of a Macromolecule by Osmometry

From $\pi = cRT = \frac{n_B RT}{V} = \frac{w_B RT}{M_B V}$:

$$M_B = \frac{w_B RT}{\pi V}$$

where w_B is mass of solute (g), V is volume of solution (L), $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$.

For macromolecules (polymers, proteins), ΔT_b and ΔT_f are extremely small and unmeasurable, but π is relatively large. Thus osmometry is the preferred method.

The **Van't Hoff equation for macromolecules** (accounting for non-ideality via second virial coefficient B'):

$$\frac{\pi}{c} = RT (1 + B'c + \dots)$$

Plotting π/c vs. c (concentration in g/L), the intercept at $c \rightarrow 0$ gives RT/M_B , and slope gives RTB' .

Worked Example: Osmotic Pressure and Freezing Point of Blood

Problem: Osmotic pressure of blood at 25°C is 7.65 atm. Find the freezing point depression. Given: $K_f = 1.86 \text{ K kg mol}^{-1}$, $R = 0.0821 \text{ L atm K}^{-1}\text{mol}^{-1}$, assuming molarity $\approx$ molality.

Step 1: Find c from Van't Hoff:

$$c = \frac{\pi}{RT} = \frac{7.65}{0.0821 \times 298} = \frac{7.65}{24.47} = 0.3127 \text{ mol L}^{-1}$$

Step 2: Since molarity $\approx$ molality:

$$\Delta T_f = K_f \times m = 1.86 \times 0.3127 = \boxed{0.582 \text{ K}}$$

So blood freezes at approximately -0.582°C .

Worked Example: Osmotic Pressure of Glucose + Sucrose

Problem: Find osmotic pressure at 25°C of a solution containing 1 g glucose ($M = 180$) and 1 g sucrose ($M = 342$) per litre.

$$c_{\text{glucose}} = \frac{1}{180} = 0.005556 \text{ mol L}^{-1}$$

$$c_{\text{sucrose}} = \frac{1}{342} = 0.002924 \text{ mol L}^{-1}$$

$$c_{\text{total}} = 0.008480 \text{ mol L}^{-1}$$

$$\pi = cRT = 0.008480 \times 0.0821 \times 298 = \boxed{0.2074 \text{ atm}}$$

Comparison of Colligative Methods for Molar Mass

Property	Formula	Best for	Sensitivity
Vapour pressure lowering	$\Delta p/p^* = x_B$	Any volatile solvent	Low
Boiling pt. elevation	$\Delta T_b = K_b m$	Small molecules	Low
Freezing pt. depression	$\Delta T_f = K_f m$	Small molecules	Medium
Osmotic pressure	$\pi = cRT$	Macromolecules	Very high

Why osmometry for macromolecules? For a polymer of $M = 100,000 \text{ g mol}^{-1}$ at 1 g/L:

$$m \approx 10^{-5} \text{ mol kg}^{-1} \implies \Delta T_f = 1.86 \times 10^{-5} \text{ K} \quad (\text{unmeasurable})$$

$$\pi = 10^{-5} \times 0.0821 \times 298 = 2.4 \times 10^{-4} \text{ atm} \approx 0.25 \text{ mmHg} \quad (\text{measurable})$$

Elevation of Boiling Point and Depression of Freezing Point: Colligative Nature via Phase Diagrams

Phase Diagram Argument

Consider the chemical potential vs. temperature diagrams for pure solvent and solution:

- $\mu_A^{\text{liq}}(\text{soln}) = \mu_A^{*,\text{liq}} + RT \ln x_A < \mu_A^{*,\text{liq}}$ at all T .
- μ_A^{solid} and μ_A^{vap} are unchanged (assuming non-volatile, insoluble solute).

The solid–liquid intersection (freezing point) moves to *lower* T because the liquid curve is shifted down. The liquid–vapour intersection (boiling point) moves to *higher* T because the liquid curve is shifted down, requiring higher T to reach the vapour curve.

Quantitative Summary

$$\begin{aligned}\Delta T_b &= K_b m \quad (\text{boiling point elevation, } K_b > 0) \\ \Delta T_f &= K_f m \quad (\text{freezing point depression, } K_f > 0) \\ K_b &= \frac{R(T_b^*)^2 M_A}{1000 \Delta_{\text{vap}} H_m^*} \\ K_f &= \frac{R(T_f^*)^2 M_A}{1000 \Delta_{\text{fus}} H_m^*}\end{aligned}$$

Osmometry and Determination of Molar Mass of Polymers

Application of Van't Hoff Equation to Polymers

The full equation used in practice is:

$$\frac{\pi}{c} = \frac{RT}{M_n} (1 + B'c)$$

where c is concentration in g/L, M_n is the **number-average molar mass**, and B' is the second virial coefficient.

Procedure:

1. Prepare several solutions of the polymer at different concentrations c .
2. Measure π for each.
3. Plot π/c vs. c .
4. Extrapolate to $c = 0$: intercept = RT/M_n .
5. Calculate $M_n = RT/(\text{intercept})$.

Significance of M_n

The number-average molar mass is:

$$M_n = \frac{\sum_i N_i M_i}{\sum_i N_i}$$

where N_i is the number of molecules of molar mass M_i . Osmometry measures M_n directly.

Worked Example: Molar Mass of Cane Sugar

Problem: A 5% cane sugar (sucrose) solution is isotonic with 8.77% urea solution. Calculate the molar mass of cane sugar. (Molar mass of urea = 60 g/mol.)

“Isotonic” means same osmotic pressure, i.e., same molar concentration:

$$c_{\text{sugar}} = c_{\text{urea}}$$

$$\frac{50}{M_{\text{sugar}}} = \frac{87.7}{60} \quad (\text{per litre, so } 5\% \text{ and } 8.77\% \text{ treated as g/100 mL} \times 10)$$

More precisely, molarity of urea = $\frac{87.7}{60} = 1.462 \text{ mol L}^{-1}$, and:

$$M_{\text{sugar}} = \frac{50}{1.462} = \boxed{342 \text{ g mol}^{-1}}$$

Abnormal Colligative Properties and Van't Hoff Factor**Electrolytes: Association and Dissociation**

For electrolytes, the experimentally observed colligative property often differs from the calculated value because:

- **Dissociation** (strong electrolytes): More particles than expected $\Rightarrow$ larger effect.
- **Association** (e.g., acetic acid in benzene): Fewer particles $\Rightarrow$ smaller effect.

Van't Hoff Factor i

$$i = \frac{\text{observed colligative property}}{\text{calculated (for non-electrolyte)}} = \frac{\text{actual moles of particles in solution}}{\text{moles of solute dissolved}}$$

For a weak electrolyte $\text{AB} \rightleftharpoons \text{A}^+ + \text{B}^-$ with degree of dissociation α :

$$i = 1 + \alpha$$

For a strong electrolyte $\text{A}_x\text{B}_y \rightarrow x\text{A}^{z+} + y\text{B}^{z-}$ (complete dissociation):

$$i = x + y$$

Relation between i and Degree of Dissociation α

For $\text{AB} \rightarrow \nu$ ions with complete dissociation:

$$i = \nu\alpha + (1 - \alpha) = 1 + (\nu - 1)\alpha$$

$$\alpha = \frac{i - 1}{\nu - 1}$$

Key Formulae Summary

Quantity	Expression
Raoult's law	$p_A = x_A p_A^*$
Vapour pressure lowering	$(p_A^* - p_A)/p_A^* = x_B$
Boiling point elevation	$\Delta T_b = K_b m$
Ebullioscopic constant	$K_b = R(T_b^*)^2 M_A / (1000 \Delta_{\text{vap}} H_m^*)$
Freezing point depression	$\Delta T_f = K_f m$
Cryoscopic constant	$K_f = R(T_f^*)^2 M_A / (1000 \Delta_{\text{fus}} H_m^*)$
Van't Hoff equation	$\pi = cRT$, or $\pi V = n_B RT$
Molar mass (osmometry)	$M_B = w_B RT / \pi V$
Molar mass (ebullioscopy)	$M_B = 1000 K_b w_B / (\Delta T_b w_A)$
Molar mass (cryoscopy)	$M_B = 1000 K_f w_B / (\Delta T_f w_A)$
Van't Hoff factor	$i = (\text{obs. prop.}) / (\text{calc. prop.})$
Degree of dissociation	$\alpha = (i - 1) / (\nu - 1)$

Additional Worked Problems

Freezing Point of Blood Sample (Full Solution)

Given: $\pi = 7.65$ atm at 25°C , $K_f = 1.86$ K kg mol⁻¹, $R = 0.0821$ L atm K⁻¹mol⁻¹.

$$c = \frac{\pi}{RT} = \frac{7.65}{0.0821 \times 298} = 0.3127 \text{ mol L}^{-1}$$

Assuming molarity $\approx$ molality :

$$\Delta T_f = K_f \times m = 1.86 \times 0.3127 = 0.582 \text{ K}$$

$$T_f(\text{blood}) = 273.15 - 0.582 = \boxed{272.57 \text{ K} \approx -0.58^\circ\text{C}}$$

Molar Mass from Freezing Point Depression

Problem: 2 g of a non-volatile solute in 50 g benzene depresses the freezing point by 0.40 K. $K_f(\text{benzene}) = 5.12 \text{ K kg mol}^{-1}$. Find molar mass.

$$m = \frac{\Delta T_f}{K_f} = \frac{0.40}{5.12} = 0.07813 \text{ mol kg}^{-1}$$

$$M_B = \frac{w_B}{m \times w_A(\text{kg})} = \frac{2}{0.07813 \times 0.050} = \frac{2}{0.003906} = \boxed{512 \text{ g mol}^{-1}}$$

Ionic Strength and Colligative Properties

For a solution of NaCl ($i = 2$) at 0.1 mol/kg:

$$\Delta T_f = iK_fm = 2 \times 1.86 \times 0.1 = 0.372 \text{ K}$$

This demonstrates that accounting for dissociation is essential for electrolyte solutions.

Exam Tips and Common Questions

Most Frequently Asked Questions from PYQs:

1. Derive Van't Hoff equation for osmotic pressure (with full thermodynamic derivation).
2. Show that elevation of boiling point / depression of freezing point is a colligative property.
3. Derive the expression for ΔT_b and ΔT_f thermodynamically.
4. What is the Van't Hoff factor? How to find degree of dissociation.
5. Determine molar mass of macromolecule by osmometry.
6. Calculate freezing point given osmotic pressure or vice versa.
7. Why is osmometry preferred for macromolecules?
8. Is osmosis a colligative property? Justify.

Key Unifying Idea: Every colligative property ultimately arises from the single equation:

$$\mu_A(\text{solution}) = \mu_A^*(\text{pure solvent}) + RT \ln x_A$$

The lowering of μ_A (since $\ln x_A < 0$) shifts every equilibrium involving the solvent: lowering vapour pressure, raising boiling point, lowering freezing point, and creating osmotic pressure. The *number* of solute particles determines x_A , making all four properties colligative.